

Software Requirements Specification (SRS)

Student OS Documentation — File 02

Purpose

Define exact software behavior for Student OS. This document converts the PRD into testable requirements.

Functional Requirements

Authentication

- Register/login/logout; secure session handling; protected routes; invalid credentials show clear errors; session expiry handled gracefully.

Dashboard

- Load personalized summary; show today's classes, next class, pending tasks, attendance, exams and notifications; all values come from application data.

Timetable

- Create/read/update/delete timetable entries; support subject, day, start/end time, room, faculty, type; validate overlapping entries where applicable.

Tasks

- Create/edit/delete/complete tasks; filter by status/date/priority/subject; due dates drive calendar and reminders.

Calendar

- Render academic and personal events; selecting a date displays relevant records; task/exam/event changes synchronize automatically.

Attendance

- Store attended/total values; calculate percentage; show subject and overall views; warn when below configured target.

Exams

- Store exam date/time/room/syllabus/status; calculate days remaining; track preparation progress.

Notes

- Create/edit/delete notes; organize by subject; support attachments/links where implemented.

Notifications

- Create and display in-app notifications; support reminder preferences and read/unread state.

Profile/Settings

- Update profile fields and preferences; protect sensitive account operations.

Non-Functional Requirements

Responsive at common desktop, tablet and mobile widths.

Accessible keyboard navigation, labels, focus states and readable contrast.

Reasonable performance for normal student-scale datasets.

Server-side authorization for protected resources.

Consistent loading, empty, success and error states.

Maintainable modular code and reusable components.

Validation & Edge Cases

Reject missing required fields, invalid dates and malformed values.

Prevent unauthorized record access.

Handle empty datasets without broken layouts.

Handle duplicate/overlapping timetable entries according to defined business rules.

Handle deleted subjects/tasks referenced by historical records safely.

Never fabricate academic data when a value is unavailable.

Acceptance

Every requirement must be testable. A feature is complete only when its happy path, validation, authorization and relevant error states work.